Algorithm to Compare Maps of a Continuous Variable at Multiple Resolutions

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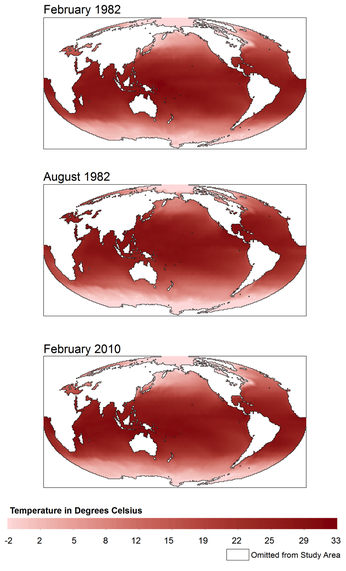
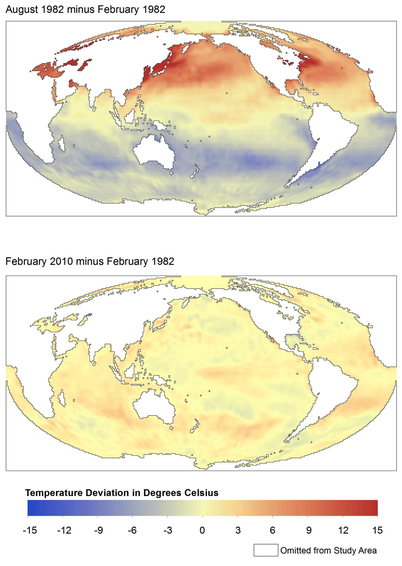
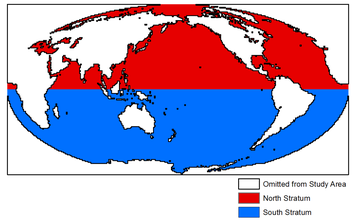


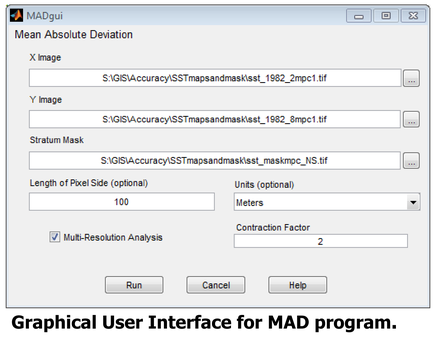
The three raster images above are Sea Surface Temperature for three time points to be used for analysis shown in the two deviation maps to the right.



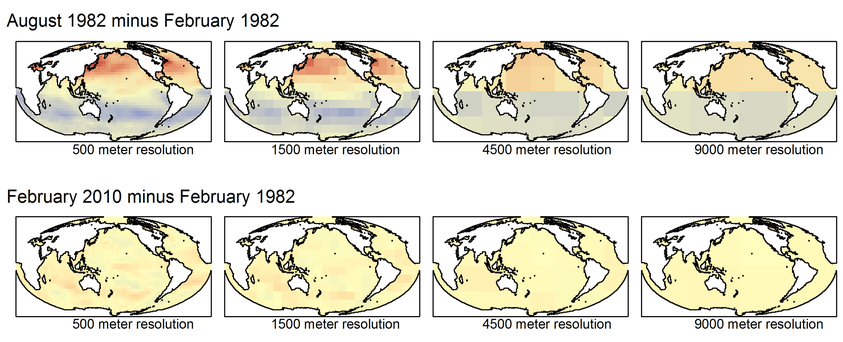
The Deviations above show warming in the northern hemisphere and cooling in the southern hemisphere from Feb 1982 to Aug 1982 , and global warming from Feb 1982 to Feb 2010.



The mask above is the Strata input used to remove all land areas and separate a Northern Stratum from a Southern Stratum at the Equator. Stratification allows one to isolate deviations among strata.



Graphical User Interface for MAD program.



Quantity Disagreement is the absolute difference between the mean of X and the mean of all Y. Stratum-level disagreement is positive when at least one stratum has a positive residual and at least one other stratum has a negative residual. Element-level component of deviation is positive when one pixel has a positive residual and another pixels within the same stratum has a negative residual. Element-level component of MAD can shrink as the spatial resolution becomes coarser.

Abstract

Mean Absolute Deviation (MAD) measures the average difference between two continuous variables that measure the same phenomenon in the same units. MAD at multiple resolutions measures how far positive deviations are from negative deviations in space. MAD is unbounded, symmetric, and is expressed in the same units as the deviations that it measures. Scientist should use this method to compare images of the same continuous variable. Applications can include measuring differences between time points, measuring differences between predicted versus observed values, and measuring differences attributable to the method of observation.

I created a computer program in MATLAB that computes MAD and its components: Quantity, Stratum, and Element. The program also outputs these components at multiple spatial resolutions with mutable spatial coarsening factors. The utility of the program is highlighted in two examples of global sea surface temperature.

Formulas

$$Dsg_{rc} = Ysg_{rc} - Xsg_{rc}$$

$$\text{Element Component at grain g} = \frac{\sum_{s=1}^{S}\sum_{r=1}^{Rg}\sum_{c=1}^{Cg}|Dsg_{rc}|}{N}$$

$$\text{Stratum Component} = \frac{\sum_{s=1}^{S}\left|\sum_{r=1}^{Rg}\sum_{c=1}^{Cg}Ds1_{rc}\right|}{N}$$

$$\text{Quantity Component} = \frac{\left|\sum_{s=1}^{S}\sum_{r=1}^{Rg}\sum_{c=1}^{Cg}Ds1_{rc}\right|}{N}$$

Xsg_{rc} is the X value in stratum s at grain g for the element in row r and column c.

Ysg_{rc} is the Y value in stratum s at grain g for the element in row r and column c.

Dsg_{rc} is the Ysg_{rc} minus Xsg_{rc} deviation in stratum s at grain g for the element in for row r and column c.

g is the coarsening factor, meaning each coarser resolution aggregates squares of g column and g rows of the finer resolution elements.

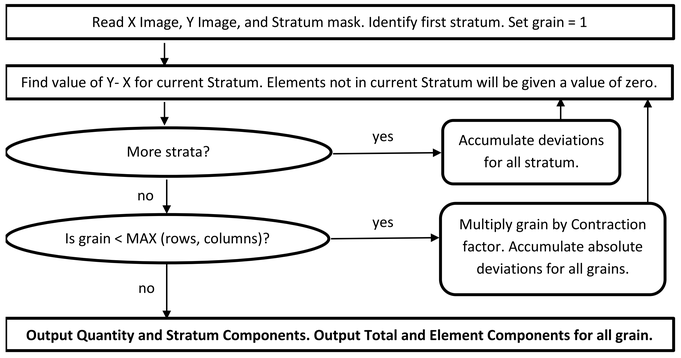
S is the number of strata.

N is the number of elements at the finest grain.

Rg is the number of rows at grain g.

Cg is the number of columns at grain g.

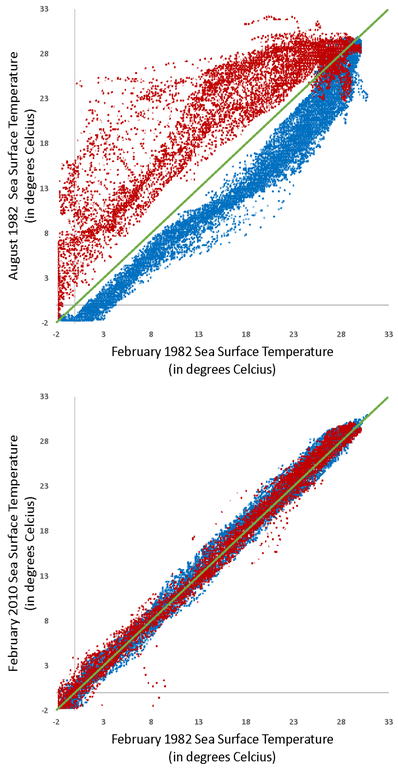
Logic



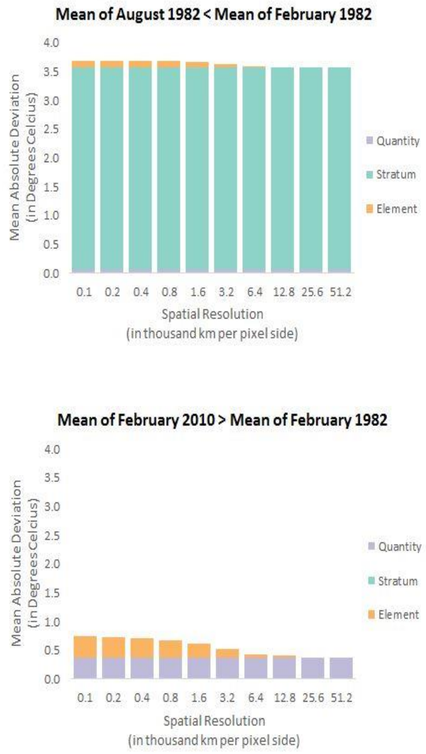
Results

The MAD program outputs the following:

- A scatterplot of the deviations in X and Y.
- A bar graph showing the changing components across grains.
- An excel sheet with the values for each component.



Scatterplot of deviations where pixels in the North Stratum are plotted in red and pixels in the South Stratum are plotted in blue. A pixel that had no change will fall on the X=Y line displayed in green.



Stacked Bar Graph displaying Element, Stratum, and Quantity components which sum to Total MAD. Pixels aggregate by an increasing coarsening factor until the entire study area is encompassed in one pixel.

Contact

I encourage you to Email me at Rchapman@clarku.edu. This free program can be downloaded from my website at chapmanrebecca.com/MAD. Please see this site for other projects, my resume, and more.

Acknowledgements

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